Student Exam Score Prediction

Linear regression with Python (Pandas, Matplotlib, Scikit-learn)

Predicting exam scores from study hours and other factors, using the Kaggle "Student Performance Factors" dataset (6,607 students).

What I did

- Cleaned the data: removed 1 row with an impossible score (101 on a 0-100 scale)
- Explored distributions and the hours vs. score relationship
- Split the data 80/20 into training and test sets (random_state=42)
- Compared four models: Hours-only linear, degree-2 polynomial, 4 numeric features, and all features (categorical columns one-hot encoded)

Results (test set)

Model	MAE	RMSE	R^2
Linear (Hours only)	2.419	3.156	0.247
Polynomial (degree 2)	2.415	3.154	0.248
Hours + Sleep + Attendance + Physical Activity	1.422	2.226	0.625
All features	0.478	2.041	0.734

Fitted equation for the Hours-only model: Score = 61.5 + 0.286 x Hours_Studied.

Key takeaways

- Hours studied alone explains about 25% of score variance. Each extra study hour is associated with about 0.29 more points.
- Polynomial regression added almost nothing, so the relationship is essentially linear.
- Adding Attendance gives the biggest jump in performance. Sleep contributes very little.
- With all features, the typical error is under 0.5 points (MAE). RMSE is much higher (2.04) because of a few students (about 0.6% of the test set) whose scores the model badly under-predicts. Nothing in their other columns explains it, so I kept them in the reported results.
- These are associations in one (likely synthetic) dataset, not proof that any factor causes higher scores.

Next steps

Cross-validation, a tree-based model for comparison, and investigating the unexplained high scorers.